



Cloud Computing with AWS

WEEK 4 – Deployment, Automation & Capstone Project

Topics Covered

- Elastic Beanstalk for automatic deployment
- AWS CloudFormation templates
- Load Balancing & Auto Scaling
- AWS Route 53 for domain management
- CI/CD with AWS CodePipeline
- Final Project Deployment and Optimization

Major Project Options

1. E-Commerce Website Deployment on AWS

- Deploy a MERN or Django app using Elastic Beanstalk.
- Store images and files in S3.
- Use RDS as database backend.

2. Serverless Web Application

- Build a frontend (HTML/React) hosted on S3.
- Backend using AWS Lambda and DynamoDB.
- API integration through API Gateway.

3 **AWS Infrastructure Automation Project**

- Use CloudFormation templates to deploy EC2, RDS, and S3 automatically.
- Implement version control with CodePipeline.

Learning Outcome

Students deploy, automate, and manage real cloud applications using various AWS services and DevOps tools.